4.1.3 Formation of ordinary differential equation:

Consider the equation $f(x, y, c_1) = 0$ --(1) where c_1 is the arbitrary constant. We form the differential equation from this equation. For this, differentiate equation (1) with respect to the independent variable occur in the equation.

Eliminate the arbitrary constant c_1 from (1) and its derivative. Then we get the required differential equation.

Suppose we have $f(x, y, c_1, c_2) = 0$. Here we have two arbitrary constants c_1 and c_2. So, find the first two successive derivatives. Eliminate c_1 and c_2 from the given function and the successive derivatives. We get the required differential equation.

Note

The order of the differential equation to be formed is equal to the number of arbitrary constants present in the equation of the family of curves.

Example 4.2

Find the differential equation of the family of straight lines $y = mx + c$ when (i) m is the arbitrary constant (ii) c is the arbitrary constant (iii) m and c both are arbitrary constants.

Solution:

(i) m is an arbitrary constant

$$y = mx + c \quad \text{...(1)}$$

Differentiating w.r. to x,

we get $\dfrac{dy}{dx} = m$ ...(2)

Now we eliminate m from (1) and (2)

For this substitute (2) in (1)

$$y = x\frac{dy}{dx} + c$$

$x\dfrac{dy}{dx} - y + c = 0$ which is the required differential equation of first order.

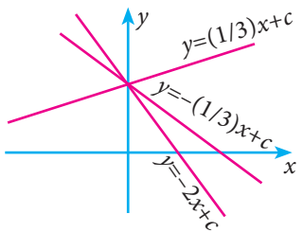


Fig. 4.1

(ii) c is an arbitrary constant

Differentiating (1), we get $\dfrac{dy}{dx} = m$

Here c is eliminated from the given equation

$\therefore \dfrac{dy}{dx} = m$ is the required differential equation.

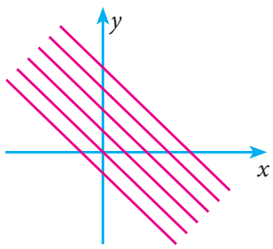


Fig. 4.2

(iii) both m and c are arbitrary constants

Since m and c are two arbitrary constants differentiating (1) twice we get

$$\frac{dy}{dx} = m$$

$$\frac{d^2y}{dx^2} = 0$$

Here m and c are eliminated from the given equation.

$\dfrac{d^2y}{dx^2} = 0$ which is the required differential equation.

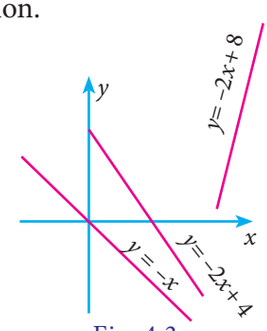


Fig. 4.3

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